

ARINZE UGWU ANSLEM

SOFTWARE ENGINEER | AI EVALUATION, TECHNICAL QA & SECURITY

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PROFESSIONAL PROFILE

Software engineer with 5+ years of experience building web and backend systems, currently developing full-stack and backend products at Nanocodes Programming Limited. Brings delivered project experience in technical QA, coding-agent evaluation, benchmark engineering, cybersecurity, fuzz-testing preparation, and smart-contract review, with advanced C++ problem-solving and practical C, C#, and Rust workflows.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, PHP (5+ years), SQL, Bash, HTML5, CSS3, and Solidity; advanced C++ code evaluation, debugging, and repository validation; practical C and C# repository evaluation; Rust fuzzing and repository workflows

Backend and Data: Node.js, Express.js, NestJS, REST, GraphQL, PostgreSQL, MySQL, MongoDB, Redis, RabbitMQ, Drizzle ORM, JWT, RBAC, and third-party integrations

Frontend: React.js, Next.js, Tailwind CSS, responsive design, component-based UI, API-driven interfaces, and browser testing

Testing and Infrastructure: pytest, unit, API, and integration testing, deterministic verifiers, regression checks, Docker, Docker Compose, Linux, WSL, AWS, Git, GitHub, clean-clone and offline builds, failure and crash reproduction, and root-cause analysis

AI Evaluation: coding-agent benchmarking, trajectory review, side-by-side response ranking, factuality research, prompt and rubric auditing, source verification, and failure-mode analysis

Security and Documentation: OWASP Top 10, API and access-control testing, fuzz-harness and seed preparation, cargo-fuzz, malformed-input testing, adversarial AI testing, Hardhat, Foundry, Slither, Microsoft Word (5+ years), Markdown, audit findings, QC notes, and technical reports

PROFESSIONAL EXPERIENCE

Full-Stack / Backend Engineer

February 2026 - Present

Nanocodes Programming Limited | Remote

- Build and maintain backend services, REST and GraphQL APIs, database schemas, and React product features using TypeScript, JavaScript, Node.js, Express.js, NestJS, PostgreSQL, and MongoDB.
- Take features from requirement clarification through API design, UI integration, validation, testing, review, debugging, documentation, and deployment preparation.
- Implement authentication and authorization with JWT and RBAC, alongside request validation, structured error handling, secure data flows, and clear service boundaries.
- Integrate third-party services and trace failures through request payloads, application logs, database behavior, environment configuration, and external dependencies.
- Improve query performance, data consistency, reliability, and maintainability through focused refactoring, regression checks, and documented fixes.

MERCOR AI EVALUATION PORTFOLIO

Software Engineering Code QA Side-by-Side Evaluator / Reviewer - Code QA SxS

July 2026

Mercor | Remote | Project paused

- Compared two blinded AI answers to developer and software-engineering questions, rating Helpfulness, Correctness, Style, and Overall Preference as separate judgments.
- Verified code, commands, configuration, APIs, Linux and shell steps, Git and CI workflows, and cloud and container behavior before rating correctness and overall quality.
- Wrote concise evidence-based rationales, passed repeated QC checks, and progressed to reviewer duties, approving defensible work and returning submissions only when material issues required correction.

Factuality and Search Quality Evaluator / Reviewer - Project Halcyon

July 2026

Mercor | Remote | Project ended

- Evaluated search results and online content for factual accuracy, error severity, and overall helpfulness across general and specialist topics.
- Established the user intent, location, and response date, separated target passages into verifiable claims, researched reputable sources, and applied the project severity framework to unsupported, contradictory, harmful, or materially false content.
- Wrote evidence-based justifications and reviewed contributor work for guideline alignment, factual support, natural wording, and internally consistent ratings.

AI Coding Agent Baseline Analyst - Imperium: VS Code vs Copilot

June 2026 - July 2026

Mercor | Remote | Project paused

- Evaluated Claude Sonnet, GPT, MAI-Code, and Raptor on identical repository-level tasks, using the same prompt and a fresh baseline attempt and container for each run.
- Inspected IDE trajectories, source changes, diffs, terminal actions, configuration, logs, official tests, and final claims across backend, systems, CLI, and full-stack repositories.
- Distinguished root-cause fixes from brittle patches, regressions, unsupported assumptions, tool misuse, scope errors, and shallow test-passing behavior while tying each submission to official tests and the required trajectory and survey evidence.
- Completed 20 approved tasks and received top-writer recognition, including several C, C++, and C# repository evaluations accepted on the first submission.

Project Ariana Reviewer and Writer / AI Prompt and Rubric Auditor

May 2026 - June 2026

Mercor | Remote

- Solved evaluation prompts before grading them, creating a reference outcome and exposing ambiguity, missing constraints, or weak assumptions in the task.
- Reviewed prompt wording, rubric answers, sampled responses, formatting, source evidence, and final-answer alignment for correctness and internal consistency.
- Worked as both writer and reviewer, producing reproducible manual verification steps, source comparisons, clear pass/fail rationales, and focused corrections without changing the original context.

Software Engineering

May 2026

Mercor | Remote | AI Agent Trajectory Evaluation

- Reviewed complete multi-turn AI agent trajectories, including the user request, plan, reasoning, tool calls, tool outputs, state changes, and final response.
- Checked whether plans covered the task, tool names and arguments were valid, constraints were followed, identifiers and numerical details were accurate, and final claims matched the execution evidence.
- Rewrote and quality-checked first-person reasoning annotations so each step responded to the latest output, explained the next action clearly, and stayed within the tools available in the task.
- Documented concrete problems in planning, tool use, execution, factual grounding, error recovery, and final-answer quality without changing the original task context.

CLI Comparisons Writer / AI Code Evaluation Analyst

April 2026

Mercor | Remote | Completed coding-agent benchmark project

- Ran the same containerized software task through five required configurations: Claude Code with Sonnet 4.6 and Opus 4.6, and Copilot CLI with Sonnet 4.6, Opus 4.6, and GPT-5.4.
- Followed the project's three-attempt limit for each configuration and documented the unresolved issue when a run did not produce a valid solution.
- Reviewed each agent's CLI actions, code changes, terminal output, and test results to determine whether the patch fixed the underlying issue and met the task requirements.
- Assessed debugging quality, instruction adherence, scope control, completeness, maintainability, and the strength of the supporting test evidence.
- Completed the final comparative survey by documenting differences in approach and verification, then ranking all five configurations from best to worst.

BENCHMARK ENGINEERING AND SECURITY EXPERIENCE**Project Fenrir - AfterQuery**

July 2026

AfterQuery | Remote | Project ended

- Evaluated private repository candidates in Rust, C/C++, and Python for architecture, input-processing surfaces, dependencies, build instructions, and security-focused fuzzing readiness.
- Prepared or assessed fuzzing harnesses, seed inputs, and repeatable environments designed to exercise parsers and other code paths with malformed or unexpected data.
- Validated Rust stable and nightly toolchains, cargo-fuzz, clean-clone builds, and offline execution in Linux and WSL while establishing repeatable crash-capture, reproduction, and triage workflows.
- Reviewed repository originality, commit history, documentation, harness connectivity, test evidence, and submission eligibility, then documented blockers and validation findings for independent review.

Software Engineering Benchmark Task Author / QA Validator - Project Pluto

2026

AfterQuery | Remote | Project ended

- Authored realistic TypeScript and Node.js benchmark repositories that required coding agents to diagnose and repair backend, API, validation, state, policy, or configuration failures.
- Wrote task instructions, reference solutions, Docker setup, and deterministic verifier suites covering seeded state, ordering, persistence, idempotency, error handling, non-mutation requirements, and edge cases.
- Completed and validated a 10-task production batch containing 144 tests with an Oracle mean of 1.000 and a no-op baseline mean of 0.000, then re-ran each package in clean environments and submitted reproducible artifacts.

Smart Contract Auditor

June 2025 - 2026

Independent Contract / Web3 | Remote | Engagement ended

- Reviewed Solidity smart contracts and DeFi business logic for access-control weaknesses, unsafe token flows, state-transition errors, missing validation, risky external interactions, and business-logic edge cases.
- Reviewed six contracts totaling approximately 1,115 lines across ERC-4337 account logic, escrow behavior, and token workflows, documenting 35 security, correctness, and maintainability findings.
- Used manual analysis, Hardhat, Foundry, and Slither to inspect contract behavior and produced reports with evidence, impact, severity reasoning, and remediation guidance.

Cybersecurity and AI Red Team Analyst

February 2025 - January 2026

Contract / Project-Based | Remote

- Conducted adversarial testing of AI systems for unsafe outputs, prompt-injection risks, exploitable behavior, weak guardrails, policy gaps, and unsupported security claims.
- Reviewed API responses, authentication assumptions, access-control logic, user-input handling, and realistic threat scenarios to identify security and abuse risks.
- Documented reproduction steps, evidence, impact, severity, and remediation recommendations for the issues identified.

Backend Security and API Testing Engineer

September 2024 - January 2025

Contract / Project-Based | Remote

- Tested backend systems and REST APIs for weaknesses in authentication, authorization, validation, data handling, error behavior, configuration, and response integrity.
- Applied OWASP Top 10 principles while using structured API requests, payload and JSON review, logs, and response analysis to reproduce and verify issues.
- Documented the evidence, likely cause, risk, and practical remediation guidance for developer review.

EARLIER AI EVALUATION AND SOFTWARE ENGINEERING EXPERIENCE**AI Systems Evaluator / Technical QA Reviewer**

March 2024 - August 2024

Contract / Project-Based Platforms | Remote

- Evaluated and ranked AI-generated text and code for factual accuracy, technical correctness, instruction adherence, completeness, and edge-case handling.
- Documented hallucinations, contradictions, missed constraints, weak assumptions, and unsupported conclusions in concise reviewer notes.

Full-Stack Web Developer

July 2023 - February 2024

AESTART | Remote

- Developed and maintained an e-commerce platform covering product catalog, pricing, inventory, cart, checkout, content, authentication, sessions, and customer-account flows.
- Integrated APIs and dynamic data flows across the frontend and backend, translating requirements into working features and diagnosing application, data, integration, and usability defects in a live product.
- Improved responsive behavior, performance, SEO, reliability, and usability while contributing to architecture and feature planning.

Frontend Web Developer

February 2022 - June 2023

Genesys Tech Hub | Remote

- Built responsive web interfaces with third-party API integrations, authentication flows, client-side state, and data-driven components.
- Worked with designers and developers to iterate on prototypes and deliver product features while testing browser behavior, responsive layouts, usability, and performance.

Backend Developer Intern

April 2021 - December 2021

ZURI | Remote

- Developed Node.js and Express backend services for the AnyLease application, including MongoDB integration, authentication, CRUD workflows, and REST endpoints.
- Used Git, debugging, request validation, and error handling to turn requirements and mockups into working backend features.

AWS Developer Intern

February 2021 - April 2021

Andela - Gauteng Program | Remote

- Completed hands-on work with AWS identity, compute, networking, database, analytics, and resource-management services.
- Applied IAM permissions, service configuration, cloud architecture, reliability, and security principles in practical exercises.

Junior Web / Backend Developer

December 2019 - January 2021

Freelance Contract and Self-Directed Projects | Remote

- Built small web and CRUD applications using Node.js, Express.js, PHP, JavaScript, HTML, CSS, MySQL, MongoDB, REST endpoints, and basic authentication.
- Handled requirements, database and request debugging, validation, Git-based version control, and technical documentation.

SELECTED TECHNICAL PROJECTS**Order Processor API***NestJS, TypeScript, PostgreSQL, Drizzle ORM, Redis, RabbitMQ, Docker Compose*

- Built a containerized order-processing API with REST endpoints, PostgreSQL persistence, Redis read-through caching and invalidation, RabbitMQ events, and a background consumer.
- Separated controllers and services, added validation, environment-based configuration, and consistent errors, then containerized the API, PostgreSQL, Redis, RabbitMQ, and pgAdmin for reproducible integration testing.

Solid FM 5Aside Football Management Platform*React.js, Next.js, TypeScript, Backend APIs, WebSocket Real-Time Updates*

- Contributed to public and administrative workflows for tournaments, teams, rosters, standings, schedules, dashboards, and live score and event controls.
- Connected responsive frontend components to backend APIs and WebSocket events and improved navigation and data presentation across desktop, mobile, public, and administrative views.

EDUCATION**Bachelor of Engineering (B.Eng.), Electronic Engineering***University of Nigeria, Nsukka | Nigeria | Second Class Upper Division*